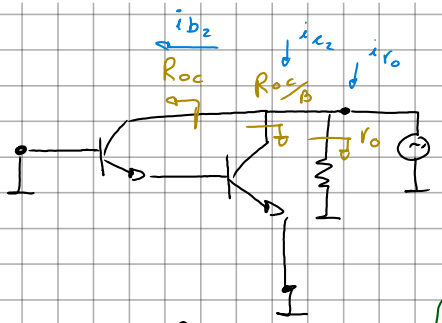


$$r_o = \frac{v_o}{i_o}$$

$$r_o^* = r_{o_2} // R_{oc} // \frac{R_{oc}}{\beta}$$

$$R_{oc} = 2 r_{o_1}$$

$$\rightarrow r_o^* = r_{o_2} // R_{oc} // \frac{R_{oc}}{\beta} \approx r_{o_2} // 2r_{o_1} // \frac{2r_{o_1}}{\beta}$$

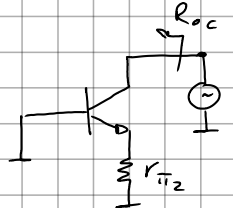
$$= r_{o_2} // 2\beta r_{o_2} // 2r_{o_2} \quad , \quad 2\beta r_{o_2} >> 2r_o > r_o$$

$$\rightarrow \boxed{r_o^* = r_{o_2} // 2r_{o_2} = \frac{2r_o \cdot r_o}{2r_o + r_o} = \frac{2}{3} r_o}$$

$$r_o = \frac{V_A}{I_{CQ}} \quad , \quad I_{CQ_2} = \beta I_{CQ_1}$$

$$\rightarrow r_{o_1} = \frac{V_A}{I_{CQ_1}} \quad , \quad r_{o_2} = \frac{V_A}{\beta I_{CQ_1}}$$

$$\rightarrow r_{o_1} = \beta r_{o_2}$$



$$R_{oc} = r_o \left(1 + \frac{\beta r_{\pi_2}}{r_{\pi_2} + r_{\pi_1}}\right) \quad , \quad \beta = r_\pi \cdot g_m$$

$$= r_o \left(1 + \frac{g_{m_1} \cdot r_{\pi_1} \cdot r_{\pi_2}}{r_{\pi_2} + r_{\pi_1}}\right)$$

$$= r_o \left(1 + g_m \left(r_{\pi_1} // r_{\pi_2}\right)\right)$$

$$\beta r_{\pi_2} = r_{\pi_1} \rightarrow r_{\pi_1} // r_{\pi_2} \approx r_{\pi_2}$$

$$\rightarrow R_{oc} = r_o (1 + g_{m_1} \cdot r_{\pi_2})$$

$$g_{m_1} \cdot r_{\pi_2} = g_{m_1} \cdot \frac{\beta}{\beta g_{m_1}} = 1$$

$$\rightarrow \boxed{R_{oc} = r_o (1 + 1) = 2 r_{o_1}}$$

$$i_{c_1} = i_{b_2} \rightarrow i_{c_2} = \beta i_{c_1}$$

$$g_m = \frac{I_{CQ}}{V_{th}}$$

$$\rightarrow g_{m_2} = \beta g_{m_1}$$

$$\text{و} \quad r_\pi = \frac{\beta}{g_m} \rightarrow r_\pi \frac{\beta}{\beta} = \frac{\beta}{g_m}$$

$$\frac{r_\pi}{\beta} = \frac{\beta}{\beta g_{m_1}}$$

$$r_{\pi_2} = \frac{r_{\pi_1}}{\beta} = \frac{\beta}{g_{m_2}}$$

$$\rightarrow \beta r_{\pi_2} = r_{\pi_1}$$